THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — MATHS · EASY LEVEL UNIT TEST SET 2

Chapters 5–7: **Arithmetic Progressions, Triangles, Coordinate Geometry** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The 1st term of an AP is 4 and the 3rd term is 10. The common difference is: 1

(a) 7 **(b)** $\frac{3}{2}$ **(c)** 6 **(d)** 3

2. Two triangles are similar if their corresponding angles are: 1

(a) in ratio 1:2 **(b)** equal **(c)** complementary **(d)** supplementary

3. The abscissa of the point $(5, 2)$ is: 1

(a) -5 **(b)** 5 **(c)** 2 **(d)** 7

4. The common difference of the AP $10, 7, 4, 1, \ldots$ is: 1

(a) -3 **(b)** 3 **(c)** -2 **(d)** -6

5. In a right triangle with legs 5 and 12, the hypotenuse is: 1

(a) 7 **(b)** 13 **(c)** 17 **(d)** 169

6. The distance of the point $(5, 12)$ from the origin is: 1

(a) 7 **(b)** 169 **(c)** 17 **(d)** 13

7. The 8th term of the AP $5, 8, 11, 14, \ldots$ is: 1

(a) 29 **(b)** 24 **(c)** 23 **(d)** 26

8. The ratio of corresponding sides of two similar triangles is 2:5. Their perimeters are in the ratio 2: 1

(a) 7 **(b)** 10 **(c)** 2 **(d)** 5

9. The midpoint of the join of $(0, 0)$ and $(8, 6)$ is: 1

(a) $(5, 3)$ **(b)** $(8, 6)$ **(c)** $(4, 4)$ **(d)** $(4, 3)$

10. The sum of the first 8 terms of $2, 4, 6, 8, \ldots$ is: 1

(a) 56 **(b)** 144 **(c)** 72 **(d)** 90

11. In $\triangle ABC$, $DE \parallel BC$, $AD = 3$, $DB = 4$, $AE = 6$. Then $EC =$ 1

(a) 12 **(b)** 8 **(c)** 4 **(d)** 6

12. The point $(-2, 5)$ lies in the: 1

(a) first quadrant **(b)** third quadrant **(c)** second quadrant **(d)** fourth quadrant

13. Which of the following sequences is an AP? 1

(a) $9, 7, 3, 1, \ldots$ **(b)** $9, 7, 6, 5, \ldots$ **(c)** $9, 7, 3, -3, \ldots$ **(d)** $9, 7, 5, 3, \ldots$

14. The areas of two similar triangles are in the ratio 9:25. The ratio of their corresponding sides is 3: 1

(a) 25 **(b)** 8 **(c)** 15 **(d)** 5

15. The coordinates of the point on the x-axis at distance 4 from the origin on the left are: 1

(a) $(4,\ 0)$ **(b)** $(0,\ -4)$ **(c)** $(-4,\ -4)$ **(d)** $(-4,\ 0)$

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Find the midpoint of the segment joining $(2,\ 6)$ and $(8,\ 10)$. 2

17. Is $2, 11, 20, 29, \ldots$ an AP? If so, find the next term. 2

18. In $\triangle ABC$, $DE \parallel BC$, $AD = 3$ cm, $DB = 4$ cm, $AE = 6$ cm. Find EC. 2

19. Name the quadrant of the point $(3, 5)$ and the coordinates of its reflection in the x-axis. 2

20. Write the first four terms of the AP with $a = 10$ and $d = -2$. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. The sides of a triangle are 6 cm, 8 cm and 10 cm. Is it a right triangle? If so, name the right angle. 3

22. Find the coordinates of the point which divides the join of $(0,\ 0)$ and $(6,\ 9)$ internally in the ratio $1 : 2$. 3

23. Find the 10th term of the AP $5, 8, 11, 14, \ldots$ 3

24. In a right triangle the two perpendicular sides are 5 cm and 12 cm. Find the hypotenuse and the perimeter. 3

OR

Find the distance between the points $(2,\ -1)$ and $(10,\ 5)$. 3

25. Find the sum of the first 8 terms of the AP $5, 10, 15, 20, \ldots$ 3

OR

In $\triangle ABC$, D and E lie on AB and AC with $DE \parallel BC$. If $AD = 3$ cm, $DB = 5$ cm and $AE = 6$ cm, find EC. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. A pole 6 m high casts a shadow 4 m long. At the same time a tower casts a shadow 30 m long. Find the height of the tower. 5

OR

Find the centroid of the triangle with vertices $(0,\ 0)$, $(6,\ 0)$ and $(0,\ 9)$. 5

27. How many terms are there in the AP $5, 10, 15, \ldots, 80$? 5

OR

A ladder 13 m long reaches a window 12 m above the ground. Find the distance of the foot of the ladder from the wall. 5

END OF QUESTION PAPER — **Total: 50 Marks**